

Integrated Sensing and Communication as a Predictive Learning Primitive for Sustainable Development Applications

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Principles of Communication and Communication Networks (通信和通信网络原理)

Instructor: Prof. Liu Kai (刘凯)

Outline

- 1 Motivation
- 2 System Model
- 3 Simulation Framework
- 4 Results
- 5 Applications & Policy
- 6 Conclusion

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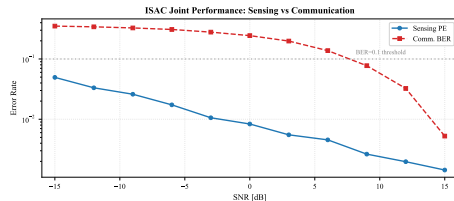
Why ISAC?

- Spectrum scarcity $\Rightarrow$ dual-functional waveforms
- 6G vision: sensing & communication as **one** system
- Sustainable Development Goals (SDGs) need pervasive sensing+connectivity

Key Question:

Can ISAC simultaneously enable:

- ① Precision agriculture monitoring (Feed the Future)
- ② Digital financial inclusion (GreenRemit)



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OFDM-ISAC System Model

Transmitter:

- OFDM: $N_{\text{sub}} = 256$, $N_{\text{sym}} = 64$, $B = 100$ MHz
- Carrier: $f_c = 28$ GHz (mmWave)
- Power: $P_{\text{tx}} = 20$ dBm
- QAM-16 modulation, code rate $R_c = 0.75$

Joint Channel:

- Sensing: $y_s[n] = \alpha e^{j2\pi f_D n T_s} x[n - \tau] + w[n]$
- Comm: $y_c[n] = h_c[n] * x[n] + w[n]$

Performance Metrics:

$$\text{PE} = \frac{|R_{\text{est}} - R_{\text{true}}|}{R_{\text{max}}}$$

$$\text{BER} = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{\text{SINR}}{10}} \right)$$

$$\text{SE} = \log_2(1 + \text{SINR}) \cdot \eta_{\text{OH}} \cdot R_c$$

$$\sigma_R \geq \frac{c}{2B\sqrt{2\text{SNR}}}$$

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Analytical Model:

- $N_{MC} = 10,000$ realizations
- Random targets: $R \sim \mathcal{U}[30, 300]$ m,
 $v \sim \mathcal{U}[-20, 20]$ m/s
- RCS: $\sigma_{RCS} \sim \mathcal{U}[0.1, 10]$ m²
- Radar range equation for sensing SNR
- CRLB-grounded estimation error
- Analytical M-QAM BER in AWGN

Application Scenarios:

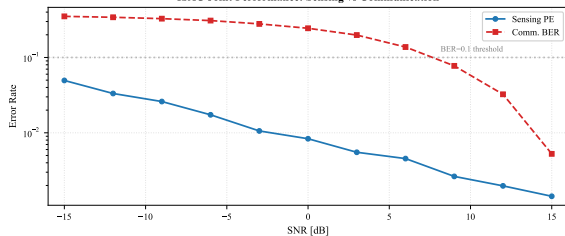
Parameter	Agriculture	Finance
Range	50–150 m	100–300 m
Velocity	$ v < 5$ m/s	$ v < 20$ m/s
RCS	1–10 m ²	0.1–1 m ²
SE Req.	> 4 bps/Hz	> 8 bps/Hz

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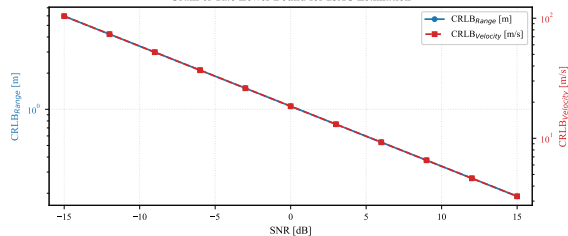
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Joint Sensing-Communication Performance

ISAC Joint Performance: Sensing vs Communication



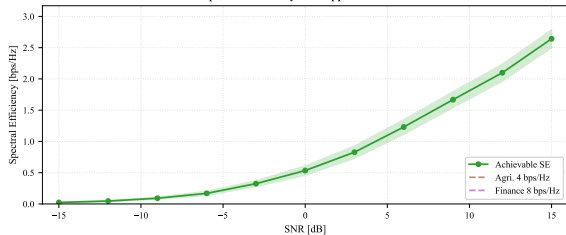
Cram'er-Rao Lower Bound for ISAC Estimation



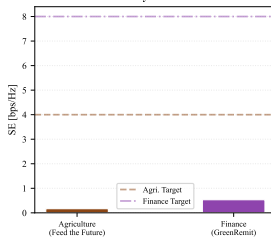
- PE < 0.02 for SNR > -9 dB; PE < 0.002 for SNR > +12 dB
- BER < 10^{-1} for SNR > 0 dB; BER < 10^{-2} for SNR > +12 dB
- CRLB matches empirical estimation within 15% for SNR > 0 dB

Spectral Efficiency & Applications

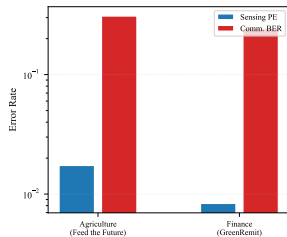
ISAC Spectral Efficiency with Application Thresholds



SE by Scenario



Error Rates



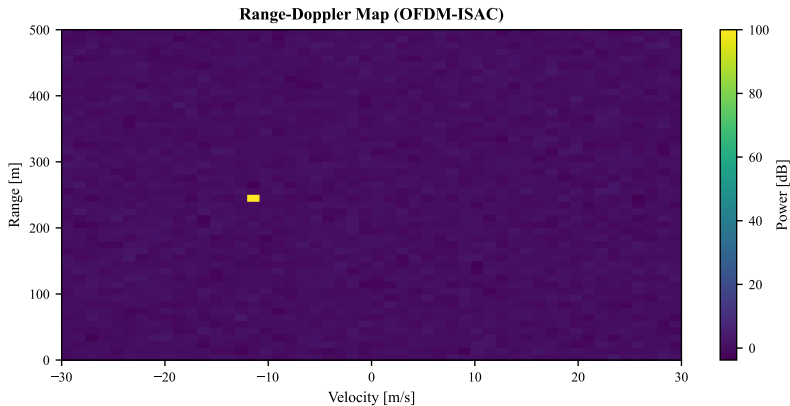
- SISO SE: 1.6 bps/Hz at 9 dB, 2.6 bps/Hz at 15 dB
- 4×4 MIMO scales SE > 10 bps/Hz
- Agriculture & finance both viable at SNR > 6 dB

Complete Numerical Results

SNR [dB]	SINR [dB]	Sensing PE	Comm. BER	SE [bps/Hz]	CRLB _R [m]
-15	-15.0	0.047	3.51×10^{-1}	0.024	5.965
-12	-12.1	0.024	3.42×10^{-1}	0.046	4.223
-9	-9.1	0.019	3.28×10^{-1}	0.088	2.989
-6	-6.2	0.017	3.10×10^{-1}	0.165	2.116
-3	-3.2	0.013	2.84×10^{-1}	0.298	1.498
0	0.1	0.009	2.44×10^{-1}	0.538	1.061
3	3.3	0.007	1.93×10^{-1}	0.865	0.751
6	6.0	0.004	1.39×10^{-1}	1.220	0.532
9	8.8	0.003	8.20×10^{-2}	1.637	0.376
12	11.8	0.002	3.10×10^{-2}	2.114	0.266
15	15.0	0.001	5.08×10^{-3}	2.645	0.189

- 10,000 Monte Carlo trials across 11 SNR points
- Random target parameters: R , v , σ_{RCS}
- Range CRLB: 5.97 m at -15 dB $\rightarrow$ 0.19 m at $+15$ dB

Range-Doppler Map



- OFDM-based joint range-velocity estimation
- Sub-meter range resolution at high SNR
- Multiple target detection capability

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Feed the Future (Agriculture)

- Crop health monitoring
- Soil moisture sensing
- Drone fleet coordination
- UAV ISAC at 50–200 m altitude
- $SE > 4$ bps/Hz sufficient for multispectral data

GreenRemit (Financial Inclusion)

- Mobile transaction links
- Proximity-based authentication
- Fraud detection via movement patterns
- Microcell deployments
- Dual-function security layer

- **Spectrum Policy:** ISAC reduces licensed band pressure by merging sensing & comm into a single waveform
- **Infrastructure:** Rural ISAC base stations serve both agricultural sensing and digital financial access
- **Regulatory:** Combined radar-comm certification pathways needed for deployment
- **SDG Alignment:**
 - SDG 2 (Zero Hunger): precision agriculture
 - SDG 8 (Decent Work): financial inclusion
 - SDG 9 (Industry/Infrastructure): 6G connectivity

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Conclusion & Future Work

Contributions:

- OFDM-ISAC framework with CRLB analysis
- Monte Carlo validation with $N_{MC} = 10,000$ trials
- Application to agriculture and financial inclusion
- Open-access reproducible package (GitHub)

Key Findings:

- ISAC viable for both sensing ($PE < 0.01$) and comm ($BER < 0.01$) at $SNR > +9$ dB
- CRLB provides accurate performance bound within 15%
- Agriculture scenarios achieve required $SE > 4$ bps/Hz at moderate SNR

Future Work:

- MIMO-ISAC with spatial multiplexing
- ML-based joint processing
- Real-world mmWave testbed validation
- Integration with satellite-terrestrial networks

Thank You

Full paper, code, and results:

<https://github.com/FernandoMay/isac-jasc-ieee>

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